Ch 4.3.3 and 4.3.4 - Multiple and Multinomial Logistic Regression

Lecture 11 - CMSE 381

Michigan State University
::
Dept of Computational Mathematics, Science & Engineering

Fri, Feb 6, 2026

Announcements

11	F	2/6	Multiple Logistic Regression / Multinomial Logistic Regression	4.3.4-5	HW #2 Due Mon 2/9	
	M	2/9	***Project Day & Review***			
	W	2/11	**Midterm #1**			
12	F	2/13	**Class not held**			
13	M	2/16	Leave one out and k-fold CV	5.1.1-3		
14	W	2/18	More k-fold CV	5.1.4-5		Q5
15	F	2/20	k-fold CV for classification	5.1.5		
16	M	2/23	Subset selection	6.1		
17	W	2/25	Shrinkage: Ridge	6.2.1		
18	F	2/27	Shrinkage: Lasso	6.2.2	HW #3 Due Sun 3/1	
	M	3/2	Spring Break			
	W	3/4	Spring Break			
	F	3/6	Spring Break			
19	M	3/9	PCA	6.3		
20	W	3/11	PCR	6.3		Q6

Announcements:

- Monday 2/9 - Project day
  - Will talk about the project
- Monday 2/9 - Review day
- Wednesday 2/11 - Exam #1
  - Bring 8.5x11 sheet of paper
  - **Handwritten** both sides
  - Anything you want on it, but must be your work
  - Must have your name and group number
  - You will turn it in
  - Calculator w/o internet

Covered in this lecture

Last Time:

- Logistic Regression

This time:

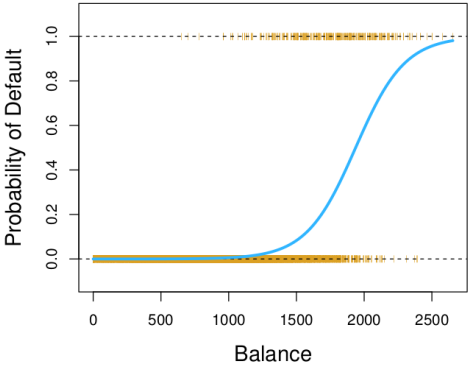
- Multiple Logistic Regression
- Multinomial Logistic Regression

Section 1

Review of Logistic Regression from last time

Logistic regression

- Assume single input $\mathtt{X}$
- Output takes values $\mathtt{Y} \in \{\mathtt{Yes}, \mathtt{No}\}$



$$p(\mathtt{X}) = \Pr(\mathtt{Y} = \mathtt{yes} \mid \mathtt{balance})$$

$$p(\mathtt{x}) = \frac{e^{\beta_0 + \beta_1 \mathtt{x}}}{1 + e^{\beta_0 + \beta_1 \mathtt{x}}}$$

- Next few slides..... how do we get this

How to get logistic function

Assume the (natural) log odds (logits) follow a linear model

$$\log\left(\frac{p(x)}{1-p(x)}\right) = \beta_0 + \beta_1 x$$

Solve for $p(x)$:

$$p(x) = \frac{e^{\beta_0+\beta_1 x}}{1+e^{\beta_0+\beta_1 x}}$$

- odds more often associated with bets "doubling down"
- human (animals) perceives doubling in physical quantity as 1 unit of perceptual change, log converts former to latter (Weber's Law)

Playing with the logistic function: desmos.com/calculator/cw1pyzzqci

Section 2

Multiple Logistic Regression

New assumption

$p \geq 1$ input variables

$$X_1, X_2, \cdots, X_p$$

Y output variable has only two levels

Multiple Logistic Regression

Multiple features:

$$p(X) = \frac{e^{\beta_0+\beta_1 X_1+\cdots+\beta_p X_p}}{1+e^{\beta_0+\beta_1 X_1+\cdots+\beta_p X_p}}$$

Equivalent to:

$$\log\left(\frac{p(X)}{1-p(X)}\right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$$

Point out the awful notation duplicating p

Example

	default	student	balance	income
0	No	No	729.526495	44361.625070
1	No	Yes	817.180407	12106.134700
2	No	No	1073.549164	31767.138950
3	No	No	529.250605	35704.493940
4	No	No	785.655883	38463.495880
5	No	Yes	919.588531	7491.558572
6	No	No	825.513331	24905.226580
7	No	Yes	808.667504	17600.451340
8	No	No	1161.057854	37468.529290
9	No	No	0.000000	29275.268290

Default data set

Predict `default` from `balance`, `student`, and `income`

$$\log\left(\frac{p(default|bal, stu, inc)}{1 - p(default|bal, stu, inc)}\right)$$
$$= \beta_0 + \beta_1\, balance + \beta_2\, student + \beta_3\, income$$

Note that student needs a dummy variable. They'll do this in the jupyter notebook, but we'll talk about both versions first

Section 3

Multinomial Logistic Regression

New assumption

$p \geq 1$ input variables

Y output variable has K levels

$$X_1, X_2, \cdots, X_p$$

Remember dummy variables?

Slide from linear regression days

`Region`:

	x_{i1}	x_{i2}
South	1	0
West	0	1
East	0	0

Create spare dummy variables:

$$x_{i1} = \begin{cases} 1 & \text{if } i\text{th person from South} \\ 0 & \text{if } i\text{th person not from South} \end{cases}$$

$$x_{i2} = \begin{cases} 1 & \text{if } i\text{th person from West} \\ 0 & \text{if } i\text{th person not from West} \end{cases}$$

Baseline is the level we're not using

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i$$

Emhpasize that we're NOT doing this. At that point we were using dummy variables in place of an input variable. Next slide we're going to be using a similar idea with output variables

Example

Predict $Y \in \{\texttt{stroke}, \texttt{overdose}, \texttt{seizure}\}$ for hospital visits based on some input(s) X

$$\Pr(Y = \texttt{stroke} \mid X = x) =$$

$$\Pr(Y = \texttt{overdose} \mid X = x) =$$

$$\Pr(Y = \texttt{seizure} \mid X = x) =$$

- We're going to figure out three numbers for any given input x, then pick the one with the highest probability
- Note that if we know two we can figure out the third

Multinomial Logistic Regression

Plan A

- Assume `Y` has K levels
- Make K (the last one) the baseline

$$\Pr(Y = k|X = x) = \frac{e^{\beta_{k0}+\beta_{k1}x_1+\cdots+\beta_{kp}x_p}}{1+\sum_{l=1}^{K-1} e^{\beta_{l0}+\beta_{l1}x_1+\cdots+\beta_{lp}x_p}}$$

$$\Pr(Y = K|X = x) = \frac{1}{1+\sum_{l=1}^{K-1} e^{\beta_{l0}+\beta_{l1}x_1+\cdots+\beta_{lp}x_p}}.$$

Example

Predict $Y \in \{\texttt{stroke}, \texttt{overdose}, \texttt{seizure}\}$ for hospital visits based on Xp

$$\Pr(Y = \texttt{stroke} \mid X = x) = \frac{\exp(\beta_{\texttt{str},0} + \beta_{\texttt{str},1}x)}{1 + \exp(\beta_{\texttt{str},0} + \beta_{\texttt{str},1}x) + \exp(\beta_{\texttt{OD},0} + \beta_{\texttt{OD},1}x)}$$

$$\Pr(Y = \texttt{overdose} \mid X = x) = \frac{\exp(\beta_{\texttt{OD},0} + \beta_{\texttt{OD},1}x)}{1 + \exp(\beta_{\texttt{str},0} + \beta_{\texttt{str},1}x) + \exp(\beta_{\texttt{OD},0} + \beta_{\texttt{OD},1}x)}$$

$$\Pr(Y = \texttt{seizure} \mid X = x) = \frac{1}{1 + \exp(\beta_{\texttt{str},0} + \beta_{\texttt{str},1}x) + \exp(\beta_{\texttt{OD},0} + \beta_{\texttt{OD},1}x)}$$

Note that using seizure is the baseline

Log odds

Calculated so that log odds between *any pair of* classes is linear.
Specifically, for $Y = k$ vs $Y = K$, we have

$$\log\left(\frac{\Pr(Y = k \mid X = x)}{\Pr(Y = K \mid X = x)}\right) = \beta_{k0} + \beta_{k1}x_1 + \cdots + \beta_{kp}x_p$$

Point out that this linear equation is the one that shows up in the top

$$\Pr(Y = k|X = x) = \frac{e^{\beta_{k0}+\beta_{k1}x_1+\cdots+\beta_{kp}x_p}}{1 + \sum_{l=1}^{K-1} e^{\beta_{l0}+\beta_{l1}x_1+\cdots+\beta_{lp}x_p}}$$

$$\Pr(Y = K|X = x) = \frac{1}{1 + \sum_{l=1}^{K-1} e^{\beta_{l0}+\beta_{l1}x_1+\cdots+\beta_{lp}x_p}}.$$

Plan B: Softmax coding

Treat all levels symmetrically

$$\Pr(Y = k|X = x) = \frac{e^{\beta_{k0}+\beta_{k1}x_1+\cdots+\beta_{kp}x_p}}{\sum_{l=1}^{K} e^{\beta_{l0}+\beta_{l1}x_1+\cdots+\beta_{lp}x_p}}.$$

Calculated so that log odds between two classes is linear

$$\log\left(\frac{\Pr(Y = k|X = x)}{\Pr(Y = k'|X = x)}\right) = (\beta_{k0} - \beta_{k'0}) + (\beta_{k1} - \beta_{k'1})x_1 + \cdots + (\beta_{kp} - \beta_{k'p})x_p.$$

Softmax coding is equivalent to the coding just described: fitted values, logodds between any pair of calsses, and other outputs are the same regardless of coding. Softmax is used exclusively in some corners of ML so good to be aware of it

Softmax example

$$\Pr(Y = \texttt{stroke} \mid X = x)$$
$$= \frac{\exp(\beta_{\texttt{str},0} + \beta_{\texttt{str},1}x)}{\exp(\beta_{\texttt{str},0} + \beta_{\texttt{str},1}x) + \exp(\beta_{\texttt{OD},0} + \beta_{\texttt{OD},1}x) + \exp(\beta_{\texttt{seiz},0} + \beta_{\texttt{seiz},1}x)}$$

$$\Pr(Y = \texttt{overdose} \mid X = x)$$
$$= \frac{\exp(\beta_{\texttt{OD},0} + \beta_{\texttt{OD},1}x)}{\exp(\beta_{\texttt{str},0} + \beta_{\texttt{str},1}x) + \exp(\beta_{\texttt{OD},0} + \beta_{\texttt{OD},1}x) + \exp(\beta_{\texttt{seiz},0} + \beta_{\texttt{seiz},1}x)}$$

$$\Pr(Y = \texttt{seizure} \mid X = x)$$
$$= \frac{\exp(\beta_{\texttt{seiz},0} + \beta_{\texttt{seiz},1}x)}{\exp(\beta_{\texttt{str},0} + \beta_{\texttt{str},1}x) + \exp(\beta_{\texttt{OD},0} + \beta_{\texttt{OD},1}x) + \exp(\beta_{\texttt{seiz},0} + \beta_{\texttt{seiz},1}x)}$$

Jupyter Notebook

Next time

11	F	2/6	Multiple Logistic Regression / Multinomial Logistic Regression	4.3.4-5	HW #2 Due Mon 2/9	
	M	2/9	***Project Day & Review***			
	W	2/11	**Midterm #1**			
12	F	2/13	**Class not held**			
13	M	2/16	Leave one out and k-fold CV	5.1.1-3		
14	W	2/18	More k-fold CV	5.1.4-5		Q5
15	F	2/20	k-fold CV for classification	5.1.5		
16	M	2/23	Subset selection	6.1		
17	W	2/25	Shrinkage: Ridge	6.2.1		
18	F	2/27	Shrinkage: Lasso	6.2.2	HW #3 Due Sun 3/1	
	M	3/2	Spring Break			
	W	3/4	Spring Break			
	F	3/6	Spring Break			
19	M	3/9	PCA	6.3		
20	W	3/11	PCR	6.3		Q6